Network Analysis and Visualization

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1 Task 1: Real-World Network Description

1.1 Real-World Networks

1.1.1 General Relativity and Quantum Cosmology Collaboration Network (ca-GrQc)

The General Relativity and Quantum Cosmology Collaboration Network (ca-GrQc) represents scientific collaborations among authors who submitted papers to the General Relativity and Quantum Cosmology (GR-QC) category of arXiv. The dataset covers submissions from January 1993 to April 2003 and is provided by the Stanford Network Analysis Project (SNAP) [1].

Each node represents an author, and an undirected edge connects two authors who co-authored at least one paper. Papers with multiple authors form fully connected subgraphs (cliques). The network contains **5,242 nodes and 14,496 edges** and is undirected and unweighted, with multiple collaborations between the same authors represented by a single edge.

The network supports the study of research communities, clustering, and connectivity in scientific collaboration.

1.1.2 Western States Power Grid

The Western States Power Grid network represents the electrical transmission infrastructure of the western United States. It is available through Mark Newman's Network Data Collection [3] and was used in Watts and Strogatz's 1998 study of small-world networks [4].

Nodes represent generators, transformers, or substations, while edges represent transmission lines connecting these facilities. The network contains **4,941 nodes and 6,594 edges** and is undirected and unweighted. It describes the physical connectivity of the power grid without including transmission capacities or electricity-flow directions.

The network supports comparisons of connectivity, clustering, and path lengths between physical infrastructure and theoretical network models.

Table 1: **Original dataset metadata.**

Network	Nodes	Edges	Directed?	Weighted?
ca-GrQc	5,242	14,496	No	No
Western States Power Grid	4,941	6,594	No	No

Sources: SNAP [1] and Mark Newman's Network Data Collection [3].

1.2 ER and BA Counterparts

For each real network with N nodes and M edges, the Erdős–Rényi (ER) model was generated using $p = 2M/[N(N-1)]$. The Barabási–Albert (BA) model used the integer attachment parameter m giving the closest available edge count. All models used random seed 42. For ca-GrQc, 12 self-loops were removed from the analysis copy before setting the model parameters, leaving 5,242 nodes and 14,484 edges.

The standard BA model could not closely match the Power Grid's 6,594 edges: $m = 1$ produced 4,940 edges (25.08% fewer), while $m = 2$ would produce 9,878 edges. Thus, $m = 1$ was selected as the closer option.

Table 2: **Reference-model parameters and generated graph sizes.**

Base network	Model	Nodes	Edges	Seed	Parameter
ca-GrQc	ER	5,242	14,176	42	$p = 0.001054405$
ca-GrQc	BA	5,242	15,717	42	$m = 3$
Power Grid	ER	4,941	6,431	42	$p = 0.000540303$
Power Grid	BA	4,941	4,940	42	$m = 1$

2 Task 2: Global Network Properties

Diameter, radius, and average eccentricity were calculated on the largest connected component (LCC) of each network. Connected-component counts and degree statistics were calculated over the full graph. Degree variance is the population variance (ddof = 0).

Table 3: **Network sizes and connectivity (full graph).**

Network	Nodes	Edges	Components	LCC nodes
ca-GrQc (real)	5,242	14,484	355	4,158
ca-GrQc (ER)	5,242	14,176	19	5,223
ca-GrQc (BA)	5,242	15,717	1	5,242
Power Grid (real)	4,941	6,594	1	4,941
Power Grid (ER)	4,941	6,431	387	4,504
Power Grid (BA)	4,941	4,940	1	4,941

Table 4: **Distance measures (LCC) and degree statistics (full graph).**

Network	Diameter	Radius	Mean eccentricity	Mean degree	Degree variance
ca-GrQc (real)	17	9	11.675	5.526	62.690
ca-GrQc (ER)	10	7	8.047	5.409	5.157
ca-GrQc (BA)	7	4	5.686	5.997	74.714
Power Grid (real)	46	23	34.541	2.669	3.209
Power Grid (ER)	20	12	14.924	2.603	2.465
Power Grid (BA)	20	10	14.390	2.000	16.740

The real ca-GrQc network has 355 components and a diameter of 17, compared with 19 components and a diameter of 10 for ER, and one component and a diameter of 7 for BA. Its degree variance (62.690) is substantially greater than ER's (5.157).

The real Power Grid is connected and has a diameter of 46, while its ER counterpart has 387 components and a diameter of 20 on its LCC. Both reference models have substantially smaller mean eccentricities than the real Power Grid.

3 Task 3: Network Distributions

Four distributions are reported for each network. Degree and local clustering use all nodes; eccentricity uses exact Task 2 results on the LCC; component size uses one observation per component, including the LCC. Table 5 gives means and population variances (ddof = 0), calculated from original observations. Probabilities are normalized by the relevant sample size. Logarithmic axes

are used where appropriate; degree plots preserve zero using a symmetric logarithmic horizontal scale. Clustering uses 20 equal-width interior bins and separate zero/one masses. Curves are descriptive, not significance tests; detailed fitting settings are documented in the supporting reproduction guide.

Table 5: **Distribution means and population variances.** Eccentricity uses LCC nodes; clustering uses all nodes; component size uses one observation per component, including the LCC.

	Eccentricity		Local clustering		Component size	
Network	Mean	Variance	Mean	Variance	Mean	Variance
ca-GrQc (real)	11.675	1.204	0.529636	0.183768	14.766	48,495.396
ca-GrQc (ER)	8.047	0.206	0.000921	0.000092	275.895	1,359,658.410
ca-GrQc (BA)	5.686	0.227	0.009484	0.002331	5,242.000	0
Power Grid (real)	34.541	13.827	0.080104	0.050260	4,941.000	0
Power Grid (ER)	14.924	1.153	0.000502	0.000281	12.767	52,257.104
Power Grid (BA)	14.390	2.701	0	0	4,941.000	0

3.1 Degree Distribution

Figures 1 and 2 show empirical degree probabilities. Real-network curves use shifted negative-binomial fits with a separate zero mass; ER curves use Binomial$(N-1, p)$ references. BA curves fit discrete power-law tails only, scaled by the empirical tail fraction. These fits do not establish a power-law hypothesis [5]. Degree means and variances appear in Table 4.

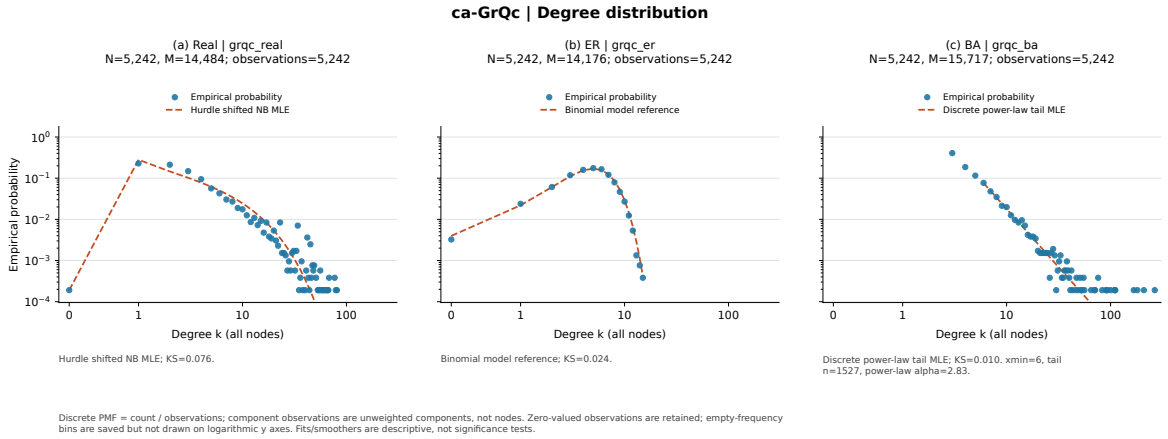


Figure 1: **Degree distributions for ca-GrQc and its ER/BA counterparts.** All nodes are included. Dashed curves show the real-network count fit, ER reference, and BA tail fit, respectively.

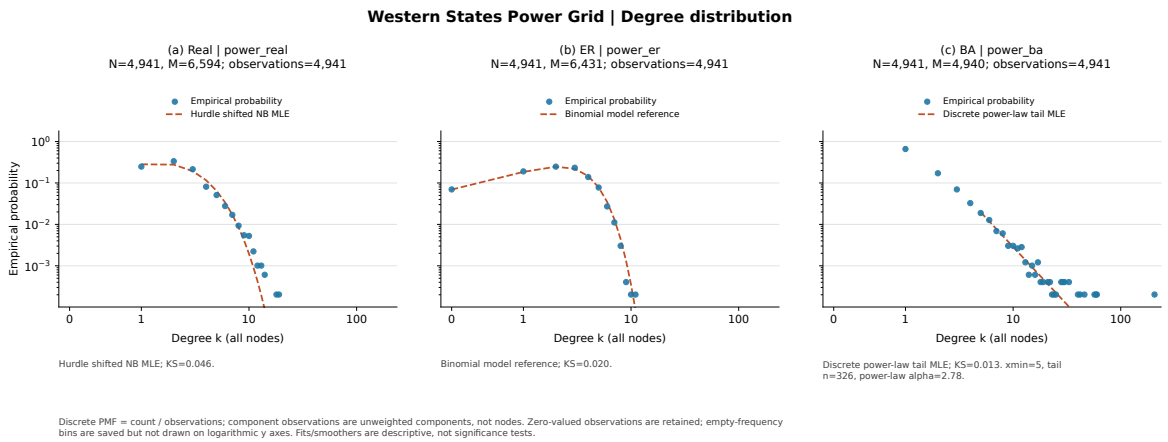


Figure 2: **Degree distributions for Power Grid and its ER/BA counterparts.** The BA curve describes degrees $k \geq 5$ only; the full empirical distribution is retained.

3.2 Longest-Shortest-Path (Eccentricity) Distribution

Eccentricity is $e(v) = \max_{u \in \mathrm{LCC}} d(v, u)$, measured in hops. Figures 3 and 4 use one observation per LCC node. Curves are bounded discrete-normal fits, except for Power Grid BA, which uses a beta-binomial fit. All curves are restricted to the observed integer support from radius to diameter.

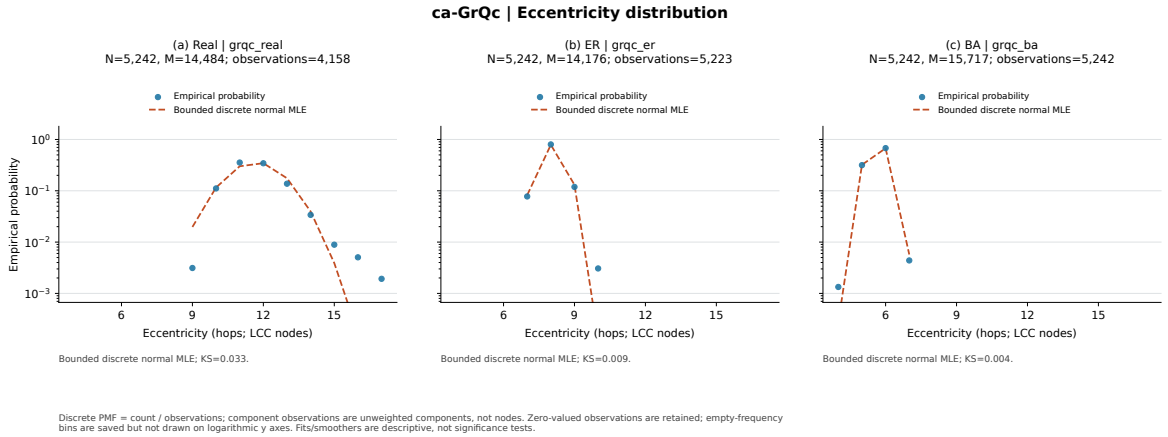


Figure 3: **Eccentricity distributions for ca-GrQc and its ER/BA counterparts.** Each observation is an LCC node. Dashed curves are bounded discrete-normal fits on the observed integer supports.

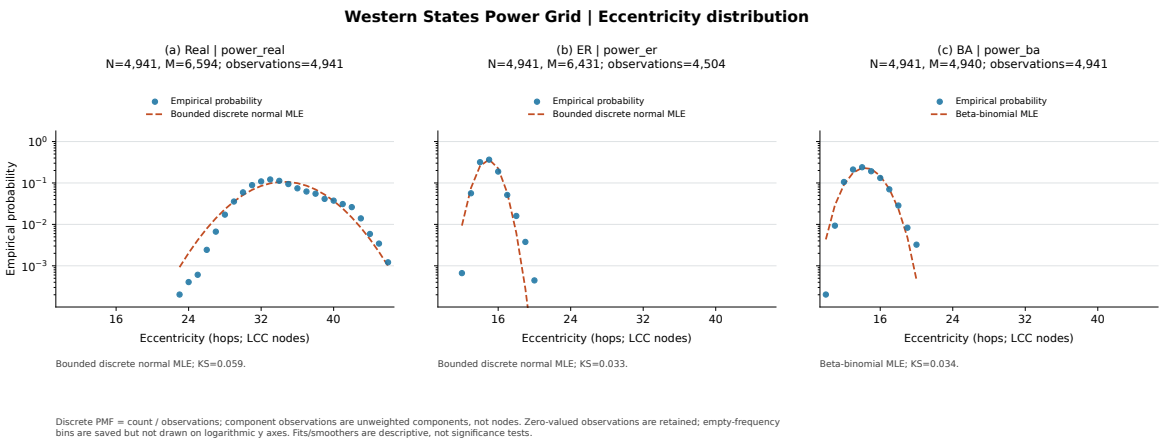


Figure 4: **Eccentricity distributions for Power Grid and its ER/BA counterparts.** Curves are bounded discrete-normal fits for real/ER and a beta-binomial fit for BA. Distances are in hops.

3.3 Clustering-Coefficient Distribution

Figures 5 and 6 include all nodes, assigning zero to degree-zero/one nodes. Four curves use reflected kernel density estimation (KDE), integrated over bins, as descriptive smoothers after Beta fits failed the selected screen. Continuous fits are N/A for Power Grid ER (eight interior observations) and BA (all-zero clustering because $m = 1$ produces a tree).

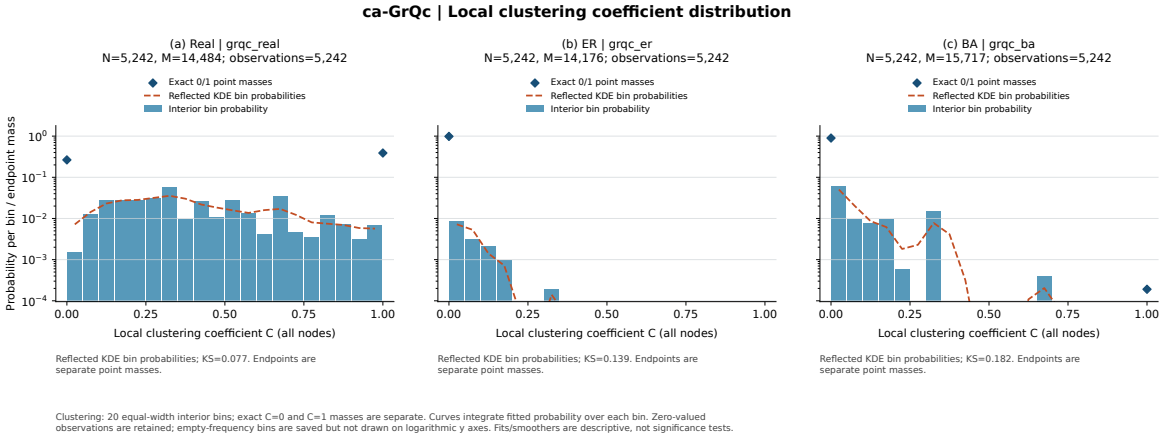


Figure 5: **Local clustering distributions for ca-GrQc and its ER/BA counterparts.** Bars show interior probabilities, diamonds show endpoint masses, and dashed curves show KDE bin probabilities.

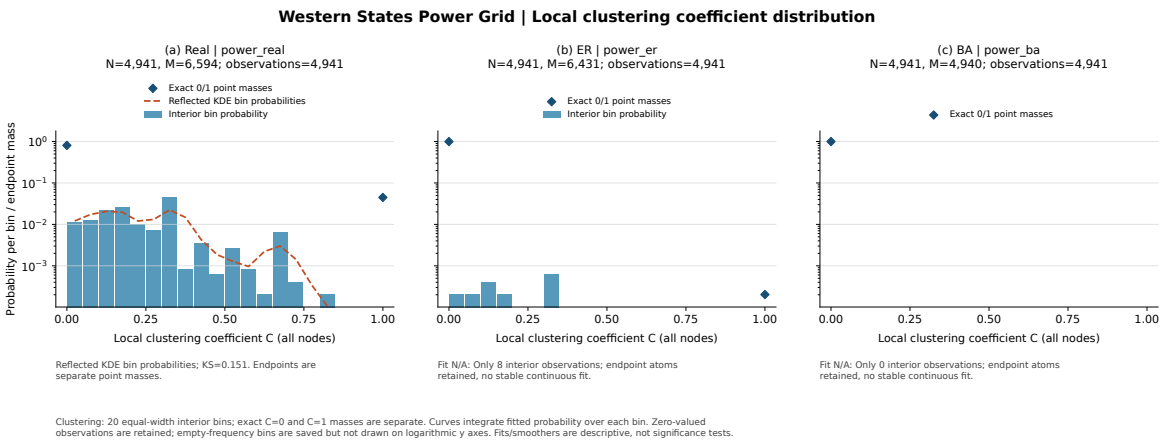


Figure 6: **Local clustering distributions for Power Grid and its ER/BA counterparts.** The real-network curve is a KDE smoother. ER/BA fits are N/A; all nodes are included.

3.4 Connected-Component Size Distribution

Figures 7 and 8 weight components equally. Power Grid ER fits non-LCC sizes with a shifted negative binomial, scaled by their component fraction. Fits are N/A for connected graphs (one observation), real ca-GrQc (failed fit screen), and ca-GrQc ER (18 non-LCC observations). The latter two follow the chosen fitting policy. All moments include the LCC.

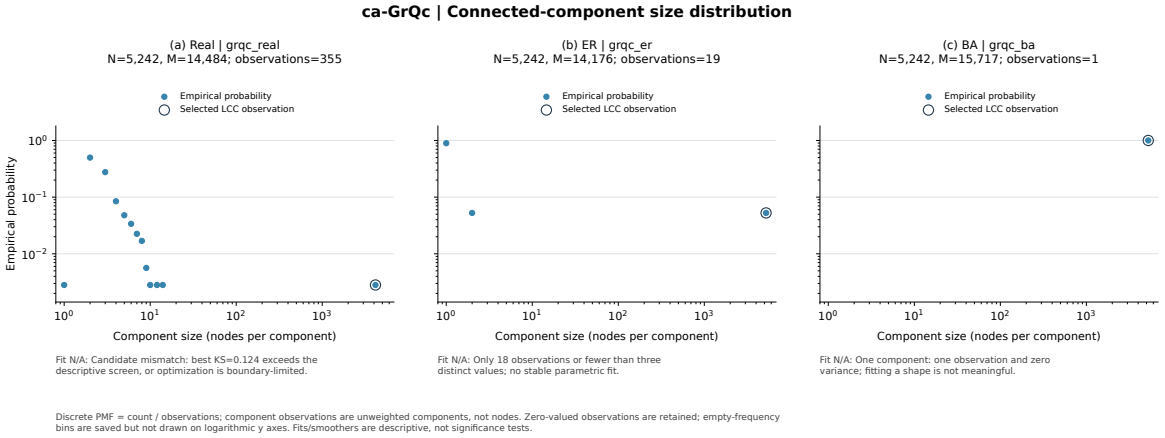


Figure 7: **Component-size distributions for ca-GrQc and its ER/BA counterparts.** Probabilities are per component, not per node. All components are shown; N/A annotations explain the omitted curves.

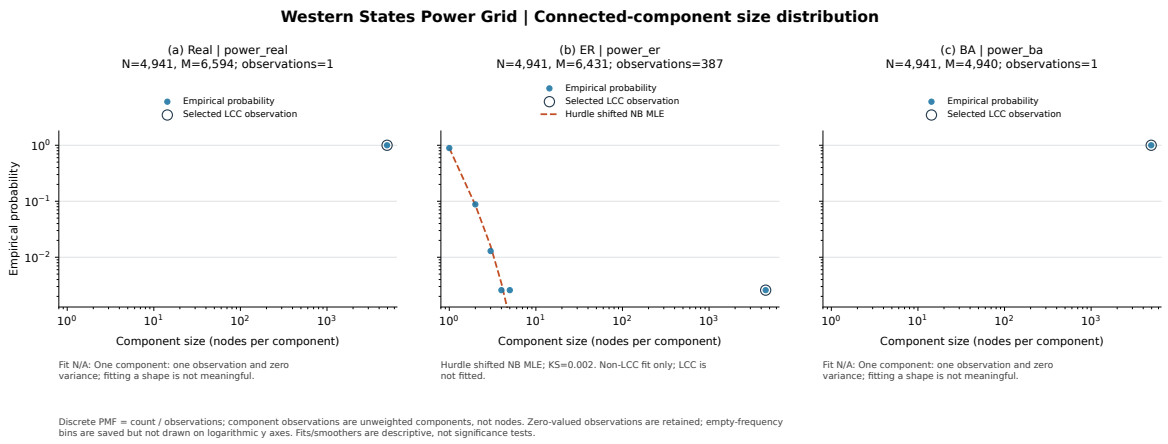


Figure 8: **Component-size distributions for Power Grid and its ER/BA counterparts.** The ER curve fits non-LCC components only; the LCC and single-component real/BA graphs remain empirical observations.

4 Task 4: Comparison and Interpretation

4.1 ca-GrQc: Real vs ER vs BA

The real ca-GrQc network has substantially greater degree heterogeneity than ER: their degree variances are 62.690 and 5.157, respectively (Table 4). BA is closer on this measure, with variance 74.714, but its maximum degree of 266 exceeds the real network's 81 (Figure 1). Thus, preferential attachment provides a more suitable description of unequal connectivity than independent random edges in this comparison, without reproducing the full observed degree distribution or establishing how the real network formed.

Neither model captures the strong local clustering. Mean clustering is 0.529636 in the real network, compared with 0.000921 for ER and 0.009484 for BA (Table 5; Figure 5). The high real-network clustering is consistent with its construction: coauthors of a multi-author paper form a clique. Neither reference model explicitly includes this group-based construction.

The models also underrepresent fragmentation and LCC distances. The real graph has 355 components, versus 19 for ER and one for BA (Table 3; Figure 7). Mean eccentricity is 11.675 in the real LCC, versus 8.047 and 5.686, respectively (Figure 3). These differences indicate that matching node counts and approximate edge counts does not reproduce component structure or distances within the largest component. BA captures degree heterogeneity more closely, but neither model jointly reproduces clustering, fragmentation, and eccentricity.

4.2 Western States Power Grid: Real vs ER vs BA

ER is closer to the Power Grid's degree variability: its variance is 2.465, compared with 3.209 for the real graph and 16.740 for BA (Table 4). Maximum degrees are 11, 19, and 206 for ER, real, and BA, respectively (Figure 2). BA therefore introduces much stronger hub dominance than observed, whereas ER better approximates the relatively narrow degree distribution.

This agreement does not extend to connectivity or distances. The real grid is connected, while ER has 387 components, including 345 isolates; BA is connected (Table 3; Figure 8). Nevertheless, both models have much smaller mean eccentricities: 14.924 for ER and 14.390 for BA, versus 34.541

for the real grid (Figure 4). ER distances describe only its LCC. The real grid's longer distances are consistent with geographically constrained infrastructure, although these measurements do not establish that explanation causally.

Mean clustering is 0.080104 in the real grid, 0.000502 in ER, and zero in BA (Figure 6). BA's zero clustering follows from its $m = 1$ tree construction. Its 25.08% edge deficit (Table 2) also confounds structural comparisons, so differences cannot be attributed solely to attachment rules. ER better approximates degree variability and BA preserves connectedness, but neither reproduces the grid's clustering and eccentricity. Across both datasets, conclusions concern single seeded realizations with unequal realized edge counts, not ensemble-level model performance.

5 Task 5: Methodology and Reproducibility

5.1 Data Sources and Preprocessing

The original ca-GrQc edge list was obtained from SNAP [1], and Power Grid from Newman's collection [3]. Raw archives and extracted files are preserved under `data/raw/`. The ca-GrQc input is `data/raw/ca-GrQc.txt/CA-GrQc.txt`; Power Grid uses `data/raw/power/power.gml`. Input SHA-256 checksums, archive-integrity checks, parsing diagnostics, and original sizes are recorded in the input-audit files under `results/tables/`.

ca-GrQc was read as a two-column integer edge list, ignoring comment and blank lines. Malformed records were checked; none were found. Reciprocal records were combined into single undirected edges without assigning weights. Its original 5,242 nodes and 14,496 edges include 12 self-loops. Only these loops were removed from the analysis copy, leaving 5,242 nodes and 14,484 edges. All vertices were retained, including node 12295, which became isolated. Power Grid was read from GML with `label=None`, preserving declared numeric node IDs. It contains no self-loops or duplicate edges, and remains at 4,941 nodes and 6,594 edges. Neither network required weight removal, imputation, or additional filtering. Both were analyzed as simple, undirected, unweighted graphs; the edge-list format cannot reveal any unlisted upstream isolates.

The LCC is the component containing the most nodes, with ties resolved by the smallest numeric node ID. Only distance-based measures use its induced subgraph; degree, clustering, and component statistics retain the full graph.

5.2 Generation Settings

Each reference model uses the analyzed real network's node count. ER uses `gnp_random_graph` with $p = 2M/[N(N-1)]$, calculated without pre-rounding, so M is the expected rather than enforced edge count. BA uses `barabasi_albert_graph` with the default initial star. Its integer m minimizes $|m(N-m) - M|$, choosing the smaller m if tied. No edges were adjusted after generation. All four samples use seed 42; Table 2 records parameters and realized sizes. Full-precision settings and edge-count differences are saved in `task2_config.json` and `results/tables/task2_model_parameters.csv`.

5.3 Software and Computation

Analysis used Python 3.12.15 in Conda environment `fons-a1`: NetworkX 3.7 for graphs, generation and clustering; NumPy 2.5.4 for moments; pandas 2.3.3 for tabular checks; SciPy 1.18.1 for fitting; Matplotlib 3.11.2 for figures; and pytest 9.1.1 for validation. The report was compiled with

MiKTeX-pdfTeX 4.27 (MiKTeX 26.5), with automatic package installation and shell execution disabled.

Exact unweighted breadth-first search was performed once per LCC node. The resulting eccentricities determine diameter, radius, and mean eccentricity and are cached for Task 3; no distance approximation was used. All variances use `ddof=0`. Clustering includes degree-zero/one nodes with coefficient zero; component-size observations are equally weighted components. Checks include degree sum $2M$, component-size sum N, LCC bounds, finite moments, deterministic model generation, and agreement between tables and cached observations. The latest independent verification passed all 55 tests.

Fits are descriptive: maximum-likelihood candidates are compared by AIC on a common sample, with a chosen KS-distance screen of 0.10 rather than a significance test. ER degree references retain the generation probability; BA degree fits use discrete power-law tails, compared with lognormal alternatives. Eccentricity candidates use observed integer bounds. Clustering retains exact zero/one masses and uses reflected Gaussian KDE with Scott bandwidth when Beta fits fail the screen. Component fits exclude the LCC, require at least 50 observations and three distinct sizes, and are scaled by the non-LCC fraction; all reported moments still include the LCC. Failed, sparse, or degenerate fits are labeled N/A. Candidate diagnostics, parameters, and limitations are retained in `task3_config.json`, the fit tables, and `TASK3_REPRODUCIBILITY.md`.

Matplotlib uses the non-interactive Agg backend. Eight three-panel composites cover all 24 distributions, with shared scales within each comparison. Figures are exported as vector PDF and 200-dpi PNG; numerical samples, probabilities, and clustering bins are saved separately. Regenerated figures require renewed visual review.

5.4 Reproduction Instructions

Use the original inputs at the paths above and the recorded package versions. From the project root, run the following PowerShell commands. The Conda path is specific to the verified workstation and may be replaced with the corresponding local installation. A project-local temporary directory avoids pytest access conflicts.

```
& .\scripts\preflight.ps1
$cx = 'C:\ProgramData\anaconda3\Scripts\conda.exe'
$env:TEMP = Join-Path $PWD 'results/cache/reproduction_tmp'
$env:TMP = $env:TEMP
New-Item -ItemType Directory -Force $env:TEMP | Out-Null
& $cx run -n fons-a1 --no-capture-output python `
  src/run_task2.py
& $cx run -n fons-a1 --no-capture-output python `
  scripts/refresh_task2_manifest.py --write
& $cx run -n fons-a1 --no-capture-output python `
  src/run_task3.py
& $cx run -n fons-a1 --no-capture-output python `
  src/run_task3.py --verify-only
& .\report\build_report.ps1
```

Task 2 audits the inputs, applies preprocessing, generates models, and computes or reuses validated exact caches. The manifest-refresh utility validates the six saved graphs, caches and numerical tables before updating the output checksums required by Task 3; it preserves the previous manifest. Task 3 then computes distributions, fits and figures without regenerating models or shortest paths. Verify before rebuilding the report: saved provenance includes the report state at calculation time. After a report edit, regenerate Task 3 outputs to record the new state before

using its verification mode; do not silently relabel stale provenance. To reproduce only Task 3 from unchanged Task 2 outputs, omit the Task 2 and manifest-refresh commands.

The reproducibility materials comprise `src/`, `scripts/`, `tests/`, both configuration files, `environment.yml`, `requirements.txt`, and the two task-specific reproduction guides. Include the source/audit metadata, six processed GraphML files, required caches, numerical CSV/JSON files, figures, and LaTeX sources with the supporting materials. Caches are Git-ignored locally, so a supplement must include them explicitly or regenerate them. The dependency file gives compatible ranges; the exact versions above and machine-readable version records identify the analyzed environment. Numerical tables retain full precision; report rounding is for presentation only.

6 Optional: Real-World Network Visualizations

6.1 ca-GrQc Visualization

Figure 9 visualizes the complete analyzed ca-GrQc coauthorship network from SNAP [1]. It retains all 5,242 authors and 14,484 edges after the previously documented removal of 12 self-loops. No visualization-only filtering is applied; disconnected components and the isolated vertex are retained.

ForceAtlas 2 in Gephi 0.11.3 uses scaling 100, strong gravity enabled, gravity 1, Barnes–Hut approximation enabled with $\theta = 1.2$, tolerance 1, and 15 threads; LinLog and overlap prevention are off. Strong gravity limits the spread of disconnected components without deleting vertices. The run was stopped manually; convergence is not asserted. Positions are for visualization and do not establish scientific similarity or geographic location.

Unweighted modularity detection at resolution 1.0, with randomization enabled, returned 390 communities and displayed modularity 0.857. Eight communities receive distinct colors; the remaining 3,435 nodes are gray to limit the number of legend categories. Community IDs and colors were recovered from the saved Gephi project, rather than inferred from visual proximity. They do not identify named research fields. The realized layout and partition are preserved because interactive randomized reruns may differ.

Final styling preserves all exported node positions and verifies every edge endpoint pair against the frozen GraphML. Node radius is $1.4 + 5\sqrt{k/81}$ on a 1,600-unit canvas; straight gray edges have width 0.65 and opacity 18%. Sparse labels, subdued edges and a restricted categorical palette reduce clutter while preserving the full graph. Reproduction instructions, validation results and parameters are recorded in `results/gephi/GRQC_REPRODUCIBILITY.md`.

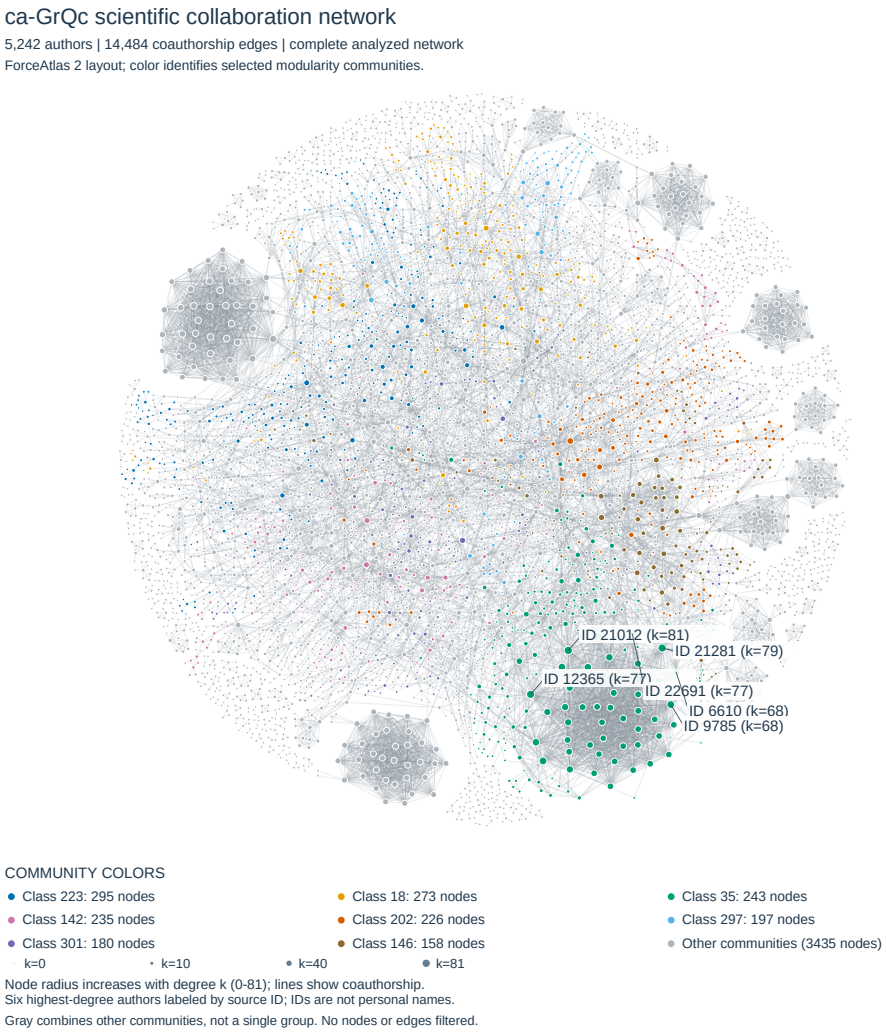


Figure 9: ca-GrQc coauthorship network. Node radius increases with degree; distinct colors identify eight selected modularity communities, while gray combines all other communities. The legend gives community IDs and node counts. Six highest-degree authors are labeled by numeric source ID (ties broken by ID); these are identifiers, not personal names. Thin gray lines show coauthorship. Source: SNAP [1].

6.2 Western States Power Grid Visualization

Figure 10 displays the complete analyzed Power Grid network from Newman's collection [3]. All 4,941 nodes and 6,594 edges are retained; no filtering or aggregation is applied. Yifan Hu layout in Gephi 0.11.3 determines positions from connectivity, not physical geography. Node radius and blue intensity encode degree, emphasizing variation in local connectivity without implying transmission capacity, power flow, or engineering criticality.

The layout used optimal distance 100, relative strength 0.2, initial step size 20, step ratio 0.95, adaptive cooling enabled, and convergence threshold 10^{-4}. The exported SVG preserves the realized coordinates, avoiding dependence on an identical rerun of the interactive layout. Final vector styling uses straight gray edges with 25% opacity and a 0.55-unit stroke on the 1,600-unit source canvas. For

$1 \le k \le 19$, node radius is $1.5 + 4.5\sqrt{(k-1)/18}$ source-canvas units; blue intensity varies linearly over the same normalized degree range. This radius mapping is a visual encoding, not a claim that area is directly proportional to degree. Labels, a degree legend, and reduced edge prominence limit clutter. The frozen GraphML, original coordinate-bearing SVG, styling/export scripts, parameter audit, and reproduction instructions are retained under `results/gephi/` and `scripts/`.

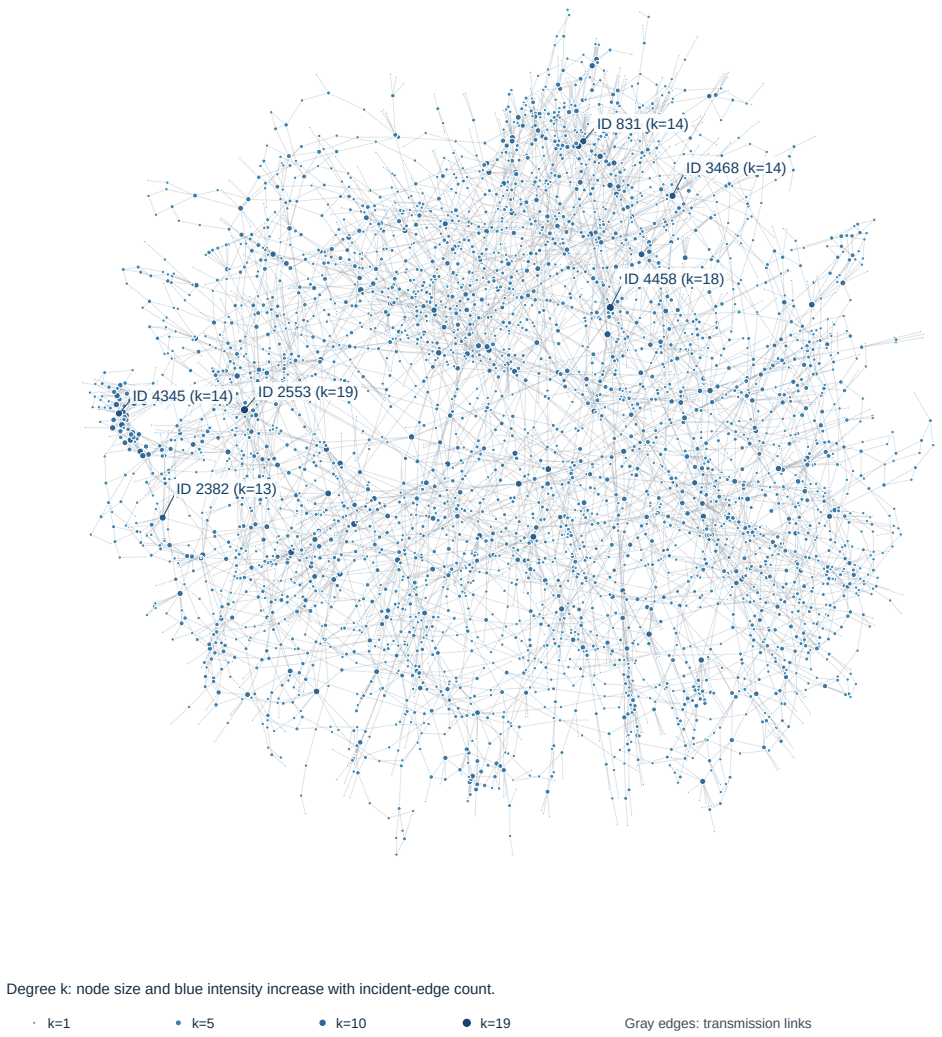


Figure 10: Western States Power Grid, shown as a complete undirected, unweighted network. Gray lines represent transmission links; node size and blue intensity increase with degree k. Callouts identify the six highest-degree nodes, with ties broken by numeric source ID. These identifiers are not facility names. Positions are topological rather than geographic. Source: Newman's network collection [3].

References

[1] Stanford Network Analysis Project (SNAP), *General Relativity and Quantum Cosmology collaboration network (ca-GrQc)*. https://snap.stanford.edu/data/ca-GrQc.html. Accessed October 17, 2026.

[2] J. Leskovec, J. Kleinberg, and C. Faloutsos, "Graph evolution: Densification and shrinking diameters," *ACM Transactions on Knowledge Discovery from Data*, vol. 1, no. 1, Article 2, 41 pages, 2007. https://doi.org/10.1145/1217299.1217301.

[3] M. E. J. Newman, *Network data: power grid of the western United States*. https://public.websites.umich.edu/~mejn/netdata/. Power Grid entry and accompanying `power.zip` documentation. Accessed October 17, 2026.

[4] D. J. Watts and S. H. Strogatz, "Collective dynamics of 'small-world' networks," *Nature*, vol. 393, pp. 440–442, 1998. https://doi.org/10.1038/30918.

[5] A. Clauset, C. R. Shalizi, and M. E. J. Newman, "Power-law distributions in empirical data," *SIAM Review*, vol. 51, no. 4, pp. 661–703, 2009. https://doi.org/10.1137/070710111.